

Create Rest Template

1-Create provider: អ្នកផ្តោត API តេទៅប្រើ

```
1 package com.demo.provider;
2
3 import org.springframework.beans.factory.annotation.Value;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.RestController;
6
7 @RestController
8 public class InstanceController {
9
10     @Value("${server.port}")
11     private String port;
12
13     private final String instanceId = java.util.UUID.randomUUID().toString();
14
15     @GetMapping("/instance-info")
16     public String getInstanceInfo() {
17         System.out.println("Request received at instance running on port: " + port);
18         return "Instance served by Port: " + port + ". Instance ID: " + instanceId;
19     }
20 }
```

```
1 spring:
2   application:
3     name: provider
4 server:
5   port: 9091
6
```

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Web WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Lombok DEVELOPER TOOLS

Java annotation library which helps to reduce boilerplate code.

Spring Boot DevTools DEVELOPER TOOLS

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

<http://localhost:9091/instance-info>

2- Create Consumer

- Create controller:

```
RestTemplateController.java ×
1 package com.demo.consumer.resttemplate;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4
5
6
7
8 @RestController
9 @RequestMapping("/api/rest-template")
10 public class RestTemplateController {
11
12     @GetMapping("/instance")
13     public String getInstance() {
14
15         RestTemplate restTemplate = new RestTemplate();
16
17         String response = restTemplate.getForObject("http://localhost:9091/instance-info",
18             String.class);
19         return response;
20     }
21 }
```

RestTemplate restTemplate = new RestTemplate(); this is we hard coding to crate object.

⇒ We will create spring bean for restTemplate.

- What we will create: configuration សម្រាប់បង្កើត Spring Bean and Service to point to the url that we want to use.

```
RestTemplateConfig.java × RestTemplateClient.java RestTemplateController.java
1 package com.demo.consumer.resttemplate;
2
3 import org.springframework.context.annotation.Bean;
4
5
6
7 @Configuration
8 public class RestTemplateConfig {
9
10     @Bean
11     public RestTemplate restTemplate() {
12         return new RestTemplate();
13     }
14
15 }
16
```

RestTemplateConfig.java RestTemplateClient.java X RestTemplateController.java

```
1 package com.demo.consumer.resttemplate;
2
3 import org.springframework.stereotype.Service;
4
5
6
7
8 @Service
9 @RequiredArgsConstructor
10 public class RestTemplateClient {
11
12     private final RestTemplate restTemplate;
13
14     private static final String PROVIDER_URL = "http://localhost:9091";
15
16     public String getInstanceInfo() {
17         return restTemplate.getForObject(PROVIDER_URL+"/instance-info", String.class);
18     }
19 }
20
21
```

RestTemplateConfig.java RestTemplateClient.java RestTemplateController.java X

```
1 package com.demo.consumer.resttemplate;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4
5
6
7
8
9
10 @RestController
11 @RequiredArgsConstructor
12 @RequestMapping("/api/rest-template")
13 public class RestTemplateController {
14
15     private final RestTemplateClient restTemplateClient;
16
17     @GetMapping("/instance")
18     public String getInstance() {
19
20         // RestTemplate restTemplate = new RestTemplate();
21         //
22         // String response = restTemplate.getForObject("http://localhost:9091/instance-info",
23         //     String.class);
24         // return response;
25
26         return restTemplateClient.getInstanceInfo();
27     }
28 }
29
```